MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, BUFFER/CONVERTER, TRUE/COMPLEMENT BUFFER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01 and 03	Inverting Hex Buffer
02 and 04	Noninverting Hex Buffer
05	Quad True/Complement Buffer
51 and 53	Inverting Hex Buffer
52 and 54	Noninverting Hex Buffer
55	Quad True/Complement Buffer

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline, (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4", flat-package)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line package)
D	F-2 (14-lead, 1/4" x 3/8", flat-package)
E	D-2 (16-lead, 1/4" x 7/8", dual-in-line package)
F	F-5 (16-lead, 1/4" x 3/8", flat-package)
X	F-1 (14-lead, 1/4" x 1/4", flat-package) except dimension "A" = 0.1 (2.54 mm) maximum
Y	F-2 (14-lead, 1/4" x 3/8", flat-package) except dimension "A" = 0.1 (2.54 mm) maximum
Z	F-5 (16-lead, 1/4" x 3/8", flat-package) except dimension "A" = 0.1 (2.54 mm) maximum

NOTES

1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines X, Y and Z only, the leads of bottom brazed ceramic packages (i.e. configuration 2 of case outlines F-1, F-2, or F-5) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e. finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.

2. For bottom or side brazed packages, case outlines X, Y and Z only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN. EG02 Marshall Space Flight Center, Alabama, 35812 by using the self-addressed Standardization Document Improvement (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Device types 01 and 02

Supply voltage range (V_{DD} - V_{SS}): $V_{CC} \leq V_{DD}$	-0.5 to +15.5 V
Output load capacitance (each output)	200 pF when V_{CC} > 10 V
Input voltage range	(V_{SS} - 0.5 V) $\leq V_I \leq$ (V_{DD} +0.5 V)

Device types 51 and 52

Supply voltage range (V_{DD} - V_{SS}): $V_{CC} \leq V_{DD}$	-0.5 to +18.0 V
Output load capacitance (each output)	200 pF when V_{CC} > 10 V
Input voltage range	(V_{SS} - 0.5 V) $\leq V_I \leq$ (V_{DD} +0.5 V)

Device types 03, 04 and 05

Supply voltage range (V_{CC} - V_{SS})	-0.5 to +15.5 V
Input voltage range	(V_{SS} - 0.5 V) $\leq V_I \leq$ (V_{DD} +0.5 V)

Device types 53, 54 and 55

Supply voltage range (V_{CC} - V_{SS})	-0.5 to +18.0 V
Input voltage range	(V_{SS} - 0.5 V) $\leq V_I \leq$ (V_{DD} +0.5 V)

All device types

Input current (each input)	±10 mA
Storage temperature range	-65°C to 175°C
Maximum power dissipation, P_D	200 mW
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case	(See MIL-M-38510, appendix C)
Junction temperature	T_J = 175°C

1.4 Recommended operating conditions.

Device types 01, 02, 03, 04, and 05

Supply voltage (V_{CC} or V_{DD} - V_{SS})	4.5 to 12.5 V dc
Input low (V_{IL}) voltage range	0-0.85 V dc @ V_{CC} or V_{DD} = 5 V; 0-2.1 V dc @ V_{CC} or V_{DD} = 12.5 V
Input high (V_{IH}) voltage range 1/	3.95-5.0 V dc @ V_{CC} or V_{DD} = 5 V; 10.0-12.5 V dc @ V_{CC} or V_{DD} = 12.5 V

Device types 51, 52, 53, 54, and 55

Supply voltage (V_{CC} or V_{DD} - V_{SS})	4.5 to 15.0 V dc
Input low (V_{IL}) voltage range	0-1.5 V dc @ V_{CC} or V_{DD} = 5 V; 0-2.0 V dc @ V_{CC} or V_{DD} = 10 V; 0-4.0 V dc @ V_{CC} or V_{DD} = 15.0 V; V_{OL} = 10% V_{DD} or V_{CC}; V_{OH} = 90% V_{DD} or V_{CC}

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Input high (V_{IH}) voltage range 1/ - - - -	3.5-5.0 V dc @ V_{CC} or V_{DD} = 5 V; 8.0-10.0 V dc @ V_{CC} or V_{DD} = 10 V; 11.0-15.0 V dc @ V_{CC} or V_{DD} = 15 V; V_{OL} = 10% V_{DD} or V_{CC}; V_{OH} = 90% V_{DD} or V_{CC}
Case operating temperature range- - - - -	-55°C to 125°C
Load capacitance- - - - - - - - - - - - - -	50 pF maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

1/ The input high voltage (V_{IH}) can exceed the power supply voltage (V_{CC}) up to the maximum rating when device types 03, 04, 53, and 54 are used for logic level conversion.

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3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. V_{IL} and V_{IH} testing requires only a summary of attribute data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and marking. All devices shall be subjected to the final electrical test in table II after part number marking, to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 37 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in tests (method 1015 of MIL-STD-883).

1. Static tests (test condition A) use circuit shown on figure 3 or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I, method 1015 for class B devices.

2. Dynamic test (test condition D) use circuit shown on figure 4, or equivalent. Ambient temperature shall be 125°C minimum. Test duration shall be in accordance with table I, method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter measurements.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_C ≤ 125°C, unless otherwise specified	Device type 1/	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD} or V_{CC}	$V_{IC(pos)}$	T_C = 25°C, V_{DD} and V_{CC} = GND, V_{SS} = Open, Output = Open, I_I = 1 mA	All		+1.5	V dc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_C = 25°C, V_{DD} and V_{CC} = Open, V_{SS} = GND, Output = Open, I_I = -1 mA	All		-6	V dc
Quiescent supply current	I_{SS}	V_{DD} and V_{CC} = 15 V dc, any combination of inputs	01-05		-750	nA dc
		V_{DD} and V_{CC} = 18 V dc, any combination of inputs	51-55		-750	nA dc
High level output voltage	V_{OH1}	V_{DD} and V_{CC} = 4.5 V dc, I_{OH} = -0.1 mA (see table III)	01-05	2.50		V dc
	V_{OH2}	V_{DD} and V_{CC} = 5 V dc, I_{OH} = -0.35 mA (see table III)	01-05	4.5		V dc
	V_{OH3}	V_{DD} and V_{CC} = 5 V dc, I_{OH} = 0 (see table III)	01-05	4.95		V dc
	V_{OH4}	V_{DD} and V_{CC} = 12.5 V dc, I_{OH} = 0 (see table III)	01-05	11.25		V dc
	V_{OH5}	V_{DD} and V_{CC} = 15 V dc, I_{OH} = 0 (see table III)	51-55	14.95		V dc
Low level output voltage	V_{OL1}	V_{DD} and V_{CC} = 5.5 V dc, I_{OL} = .23 mA (see table III)	01-05		0.5	V dc
	V_{OL2}	V_{DD} and V_{CC} = 5 V dc, I_{OL} = 2.1 mA (see table III)	01-05		0.5	V dc
	V_{OL3}	V_{DD} and V_{CC} = 5 V dc, I_{OL} = 0 (see table III)	01-05		0.05	V dc
	V_{OL4}	V_{DD} and V_{CC} = 12.5 V dc, I_{OL} = 0 (see table III)	01-05		1.25	V dc
	V_{OL5}	V_{DD} and V_{CC} = 15 V dc, I_{OL} = 0 (see table III)	51-55		0.05	V dc

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_C ≤ 125°C, unless otherwise specified	Device type 1/	Limits Min	Limits Max	Unit
Input high voltage	V_{IH1}	V_{DD} and V_{CC} = 5 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53	4.0		V dc
			52,54,55	3.5		
	V_{IH2}	V_{DD} and V_{CC} = 10 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53	8.0		V dc
			52,54,55	7.0		
	V_{IH3}	V_{DD} and V_{CC} = 15 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53	12.0		V dc
			52,54,55	11.0		
Input low voltage	V_{IL1}	V_{DD} and V_{CC} = 5 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53		1.0	V dc
			52,54,55		1.5	
	V_{IL2}	V_{DD} and V_{CC} = 10 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53		2.0	V dc
			52,54,55		3.0	
	V_{IL3}	V_{DD} and V_{CC} = 15 V dc, V_O = (see table III), $\|I_O\| \leq 1$ µA	51, 53		2.5	V dc
			52,54,55		4.0	
Output low (sink) current	I_{OL1}	V_{DD} and V_{CC} = 5 V dc, V_{IN} = (see table III), V_{OL} = 0.4 V dc	51-54	2.2		mA dc
			55	1.2		
	I_{OL2}	V_{DD} and V_{CC} = 15 V dc, V_{IN} = (see table III), V_{OL} = 1.5 V dc	51-54	17.0		mA dc
			55	13.0		
Output high (source) current	I_{OH1}	V_{DD} and V_{CC} = 5 V dc, V_{IN} = (see table III), V_{OH} = 4.6 V dc	51-54	-0.36		mA dc
			55	-1.2		
	I_{OH2}	V_{DD} and V_{CC} = 15 V dc, V_{IN} = (see table III), V_{OH} = 13.5 V dc	51-54	-2.4		mA dc
			55	-13.0		
Input leakage current	I_{IH}	V_{DD} and V_{CC} = 15 V dc	01-05		45	nA
		V_{DD} and V_{CC} = 18 V dc	51-55			
	I_{IL}	V_{DD} and V_{CC} = 15 V dc	01-05		-45	nA
		V_{DD} and V_{CC} = 18 V dc	51-55			
Input test voltage	V_{ZAP}	C_1 = 100 pF, R_Z = 1.5 kΩ (see 4.5.3)	All	400		V

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V, -55°C ≤ T_C ≤ 125°C, unless otherwise specified	Device type 1/	Limits Min	Limits Max	Unit
Input capacitance	C_i	V_{DD} and V_{CC} = 0 V dc f = 1 MHz T_C = 25°C	01,51 03,53 05,55		20	pF
			02,52 04,54		12	
Propagation delay high to low level	t_{PHL}	V_{DD} and V_{CC} = 5 V dc, C_L = 50 pF (see figure 5)	51, 52, 01, 02	7	150	ns
			54, 04	11	225	ns
			55, 05	8	172	ns
			53, 03	2	225	ns
Propagation delay low to high level	t_{PLH}	V_{DD} and V_{CC} = 5 V dc, C_L = 50 pF (see figure 5)	51, 52, 01, 02	10	210	ns
			54, 04	17	345	ns
			55, 05	9	188	ns
			53, 03	8	345	ns
Transition times high to low level	t_{THL}	V_{DD} and V_{CC} = 5 V dc, C_L = 50 pF (see figure 5)	51, 52, 01, 02	20	90	ns
			53, 54, 03, 04	5	105	ns
			55, 05	8	165	ns
Transition times low to high level	t_{TLH}	V_{DD} and V_{CC} = 5 V dc, C_L = 50 pF (see figure 5)	51, 52, 01, 02, 53, 54, 03, 04	20	405	ns
			55, 05	9	180	ns

1/ Device types 01, 51, 52, and 02 have both V_{CC} and V_{DD} terminals. Device types 03, 53, 54, and 04 have only a V_{CC} terminal. Device type 05 and 55 have only a V_{DD} terminal.

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TABLE II. Burn-in and electrical test requirements for device classification.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 1/				Class B device 1/			
		Ref. par.	Table 2/ III subgroups	Table 3/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III subgroups	Table 3/ IV delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1				1		
2	Static burn-in I method 1015	4.2b 4.5.2			3				
3	Same as line 1		1	Δ					
4	Static burn-in II method 1015	4.2b 4.5.2			3	4.2b 4.5.2	4/		3
5	Same as line 1		1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in method 1015	4.2b 4.5.2			4				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters method 5004		1*,2,3,9				1*,2,3,9		
9	Group A end point electrical parameters method 5005	4.4.1	1,2,3,4,9, 10,11			4.4.1	1,2,3,4,9		
10	Group B end point electrical parameters method 5005	4.4.2	1,2,3,9, 10,11	Δ		N.A.			
11	Group C end point electrical parameters method 5005					4.4.3	1,2,3	Δ	
12	Additional group C tests, method 5005					4.4.3c	10,11		
13	Group D end point electrical parameters method 5005	4.4.4	1,2,3			4.4.4	1,2,3		

1/ Blank spaces indicate tests are not applicable.
2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
3/ (Δ) indicates delta limit shall be required on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.
4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot; the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, if a manufacturer qualifies to a 51-55 device type which is manufactured identically to a 01-05 device type on this specification, then the 01-05 device type may be part I qualified by conducting only group A electrical tests and any electricals specified as additional group C subgroups and submitting data in accordance with MIL-M-38510, appendix D (i.e. groups A, B, C, D, and E tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

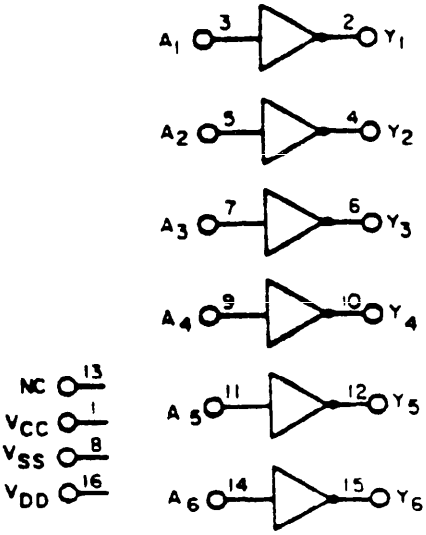
a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 4, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be performed on each inspection lot for initially qualified device types 01, 02, and 05, and measured only for initial qualification and after process or design changes for initially qualified device types 03, 04, and 51-55. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

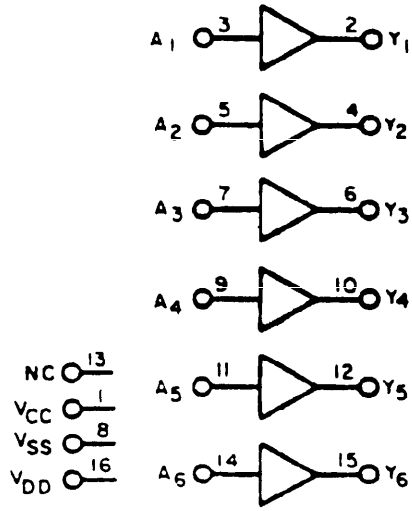
c. End-point electrical parameters shall be as specified in table II herein and shall consist of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspection, and shall consist of test specified in table IV herein.

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Device types 01, 03, 51 and 53



Device types 02, 04, 52 and 54



NOTE: Terminal 16 is not connected for device types 03, 04, 53 and 54.

Device types 05 and 55

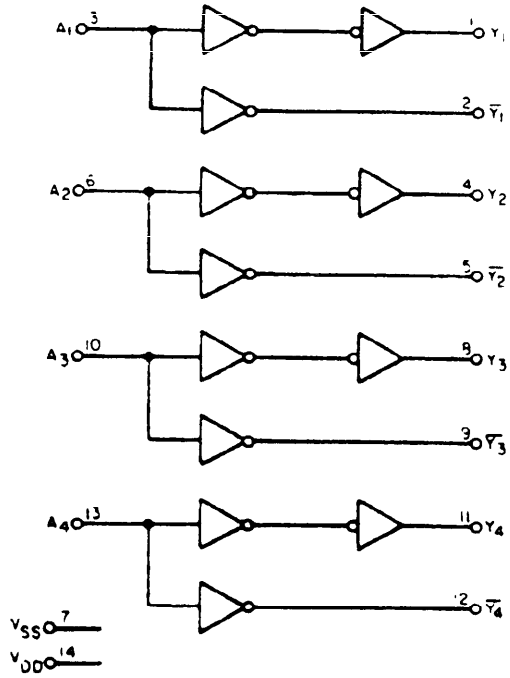


FIGURE 1. Logic diagrams and terminal connections.

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Device types 01, 03, 51 and 53

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic: $Y = \overline{A}$

Device types 02, 04, 52 and 54

Truth table each gate	
Input	Output
A	Y
L	L
H	H

Positive logic: Y = A

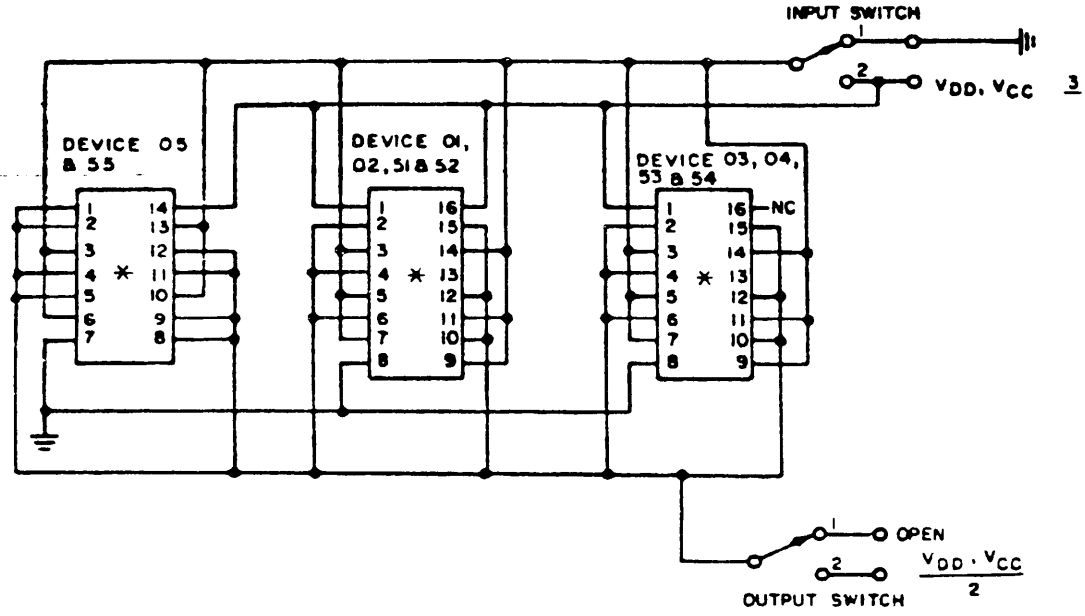
Device types 05 and 55

Truth table each gate		
Input	Output	
A	Y	$\overline{Y}$
L	L	H
H	H	L

Positive logic: Y = A

FIGURE 2. Truth tables and logic equations.

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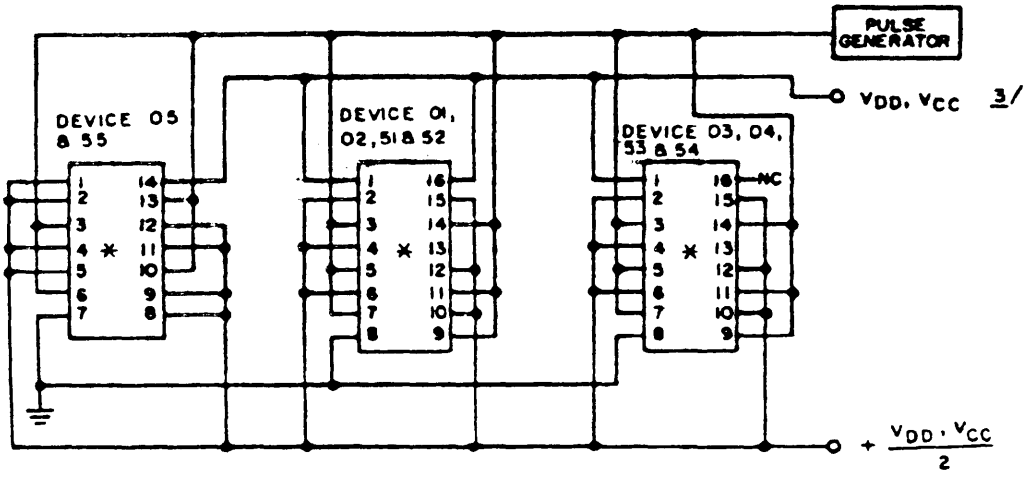


* SEE NOTE 3

NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01, 02, 03, 04 and 05.
   V_{DD} = 15.0 V minimum, 18.0 V maximum for device types 51, 52, 53, 54 and 55.
   $V_{DD/2} = V_{DD/2} \pm 1.0$ V for all devices.
   V_{SS} = 0.0 V.

FIGURE 3. Static burn-in test circuits.

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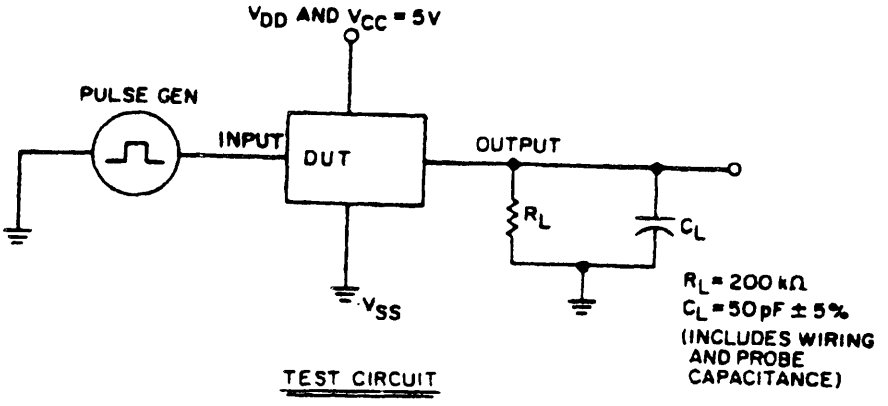


* SEE NOTE 1.

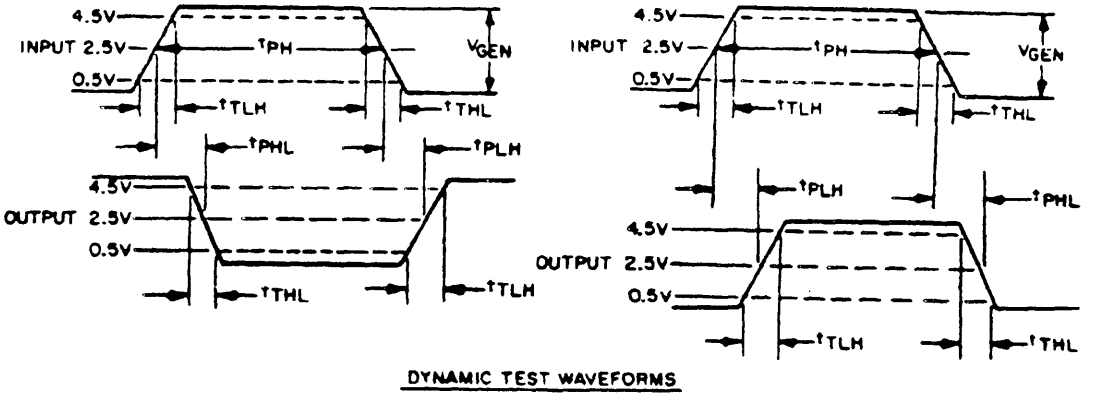
NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01, 02, 03, 04 and 05.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51, 52, 53, 54 and 55.
   $V_{DD/2}$ = $V_{DD}/2$ ±1.0 V.
   V_{SS} = 0.0 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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TEST CIRCUIT



DYNAMIC TEST WAVEFORMS

NOTES:
1. The pulse generator has the following characteristics: $V_{gen} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL} = t_{TLH} = 10 \pm 2$ ns and PRR = 200 kHz.
2. See table III for complete terminal conditions.

FIGURE 5. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device types 01 and 03.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$	See 4/	1 2 3 4 5 6	GND		1 mA		1 mA		1 mA		1 mA		1 mA			1 mA		GND	A1 A2 A3 A4 A5 A6		1.5					Vdc
$V_{IC(neg)}$		7 8 9 10 11 12			-1 mA		-1 mA		-1 mA	GND	-1 mA		-1 mA			-1 mA			A1 A2 A3 A4 A5 A6		-6					
I_{SS} I_{SS}	3005 see 5/	13 14	15.0 V 15.0 V		15.0 V GND		15.0 V GND		15.0 V GND		15.0 V GND		15.0 V GND			15.0 V GND		15.0 V 15.0 V	V_{SS} V_{SS}		-75 -75		-750 -750			nA nA
V_{OH1}	3006	15 16 17 18 19 20	4.5 V	I_{OH1}	V_{IL1}	I_{OH1}	V_{IL1}	I_{OH1}	V_{IL1}		V_{IL1}	I_{OH1}	V_{IL1}	I_{OH1}		V_{IL1}	I_{OH1}	4.5 V	Y1 Y2 Y3 Y4 Y5 Y6	2.5		2.5		2.5		Vdc
V_{OH2}		21 22 23 24 25 26	5.0 V	I_{OH2}	V_{IL1}	I_{OH2}	V_{IL1}	I_{OH2}	V_{IL1}		V_{IL1}	I_{OH2}	V_{IL1}	I_{OH2}		V_{IL1}	I_{OH2}	5.0 V	Y1 Y2 Y3 Y4 Y5 Y6	4.5		4.5		4.5		
V_{OH3}		27 28 29 30 31 32			V_{IL1}		V_{IL1}		V_{IL1}		V_{IL1}		V_{IL1}			V_{IL1}			Y1 Y2 Y3 Y4 Y5 Y6	4.95		4.95		4.95		
V_{OH4}		33 34 35 36 37 38	12.5 V		V_{IL2}		V_{IL2}		V_{IL2}		V_{IL2}		V_{IL2}			V_{IL2}		12.5 V	Y1 Y2 Y3 Y4 Y5 Y6	11.25		11.25		11.25		
V_{OL1}	3007	39 40 41 42 43 44	5.5 V	I_{OL1}	V_{IH1}	I_{OL1}	V_{IH1}	I_{OL1}	V_{IH1}		V_{IH1}	I_{OL1}	V_{IH1}	I_{OL1}		V_{IH1}	I_{OL1}	5.5 V	Y1 Y2 Y3 Y4 Y5 Y6		0.5		0.5		0.5	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device types 01 and 03 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/																	Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/			Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}			Min	Max	Min	Max	Min	Max	
V_{OL2}	3007	45 46 47 48 49 50	5.0 V	I_{OL2}	V_{IH1}	I_{OL2}	V_{IH1}	I_{OL2}	V_{IH1}	GND	V_{IH1}	I_{OL2}	V_{IH1}	I_{OL2}		V_{IH1}	I_{OL2}	5.0 V	Y1 Y2 Y3 Y4 Y5 Y6			0.5		0.5		0.5	Vdc
V_{OL3}		51 52 53 54 55 56			V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}			V_{IH1}			Y1 Y2 Y3 Y4 Y5 Y6		0.05		0.05		0.05	
V_{OL4}		57 58 59 60 61 62	12.5 V		V_{IH2}		V_{IH2}		V_{IH2}		V_{IH2}		V_{IH2}			V_{IH2}		12.5 V	Y1 Y2 Y3 Y4 Y5 Y6		1.25		1.25		1.25	
I_{IH1} 6/	3010	63	15.0 V		15.0 V		15.0 V		15.0 V		15.0 V		15.0 V			15.0 V		15.0 V	All inputs together		6					nA
I_{IH2}		64 65 66 67 68 69			15.0 V GND		GND 15.0 V GND		GND GND 15.0 V GND		GND GND GND 15.0 V GND		GND 15.0 V GND			GND 15.0 V			A1 A2 A3 A4 A5 A6		1		45			
I_{IL1} 6/	3009	70														GND			All inputs together		-6					
I_{IL2}		71 72 73 74 75 76																	A1 A2 A3 A4 A5 A6		-1		-45			
																				Subgroup 4 T_C = 25°C							
																				Min	Max						
C_1	3012	77 78 79 80 81 82	GND		A 7/		A 7/		A 7/	GND	A 7/		A 7/			A 7/		GND	A1 A2 A3 A4 A5 A6		20					pF

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device types 01 and 03 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/															Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max 8/	Min	Max 8/	Min	Max 8/	
t_{PHL}	3003 Fig. 5	83 84 85 86 87 88	5.0 V	OUT	IN	OUT	IN	OUT	IN	GND	IN	OUT	IN	OUT		IN	OUT	5.0 V	A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A5 to Y5 A6 to Y6	7/2	100/ 150	7/2	150/ 225	7/2	100/ 150	ns
t_{PLH}		89 90 91 92 93 94		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A5 to Y5 A6 to Y6	10/8	140/ 230	10/8	210/ 345	10/8	140/ 230	
t_{THL}	3004 Fig. 5	95 96 97 98 99 100		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1 Y2 Y3 Y4 Y5 Y6	20/5	60/70	20/5	90/ 105	20/5	60/70	
t_{TLH}		101 102 103 104 105 106		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1 Y2 Y3 Y4 Y5 Y6	20	270	20	405	20	270	

See footnotes at end of device type 05.

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TABLE III. Group A inspection fordevice types 02 and 04.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$	See 4/	1 2 3 4 5 6	GND		1 mA		1 mA		1 mA		1 mA		1 mA			1 mA		GND	A1 A2 A3 A4 A5 A6		1.5					Vdc
$V_{IC(neg)}$		7 8 9 10 11 12			-1 mA		-1 mA		-1 mA	GND	-1 mA		-1 mA			-1 mA			A1 A2 A3 A4 A5 A6		-6					
I_{SS} I_{SS}	3005 see 5/	13 14	15.0 V 15.0 V		15.0 V GND		15.0 V GND		15.0 V GND		15.0 V GND		15.0 V GND			15.0 V GND		15.0 V 15.0 V	V_{SS} V_{SS}		-75 -75		-750 -750			nA nA
V_{OH1}	3006	15 16 17 18 19 20	4.5 V	I_{OH1}	V_{IH1}	I_{OH1}	V_{IH1}	I_{OH1}	V_{IH1}		V_{IH1}	I_{OH1}	V_{IH1}	I_{OH1}		V_{IH1}	I_{OH1}	4.5 V	Y1 Y2 Y3 Y4 Y5 Y6	2.5		2.5		2.5		Vdc
V_{OH2}		21 22 23 24 25 26	5.0 V	I_{OH2}	V_{IH1}	I_{OH2}	V_{IH1}	I_{OH2}	V_{IH1}		V_{IH1}	I_{OH2}	V_{IH1}	I_{OH2}		V_{IH1}	I_{OH2}	5.0 V	Y1 Y2 Y3 Y4 Y5 Y6	4.5		4.5		4.5		
V_{OH3}		27 28 29 30 31 32			V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}			V_{IH1}			Y1 Y2 Y3 Y4 Y5 Y6	4.95		4.95		4.95		
V_{OH4}		33 34 35 36 37 38	12.5 V		V_{IH2}		V_{IH2}		V_{IH2}		V_{IH2}		V_{IH2}			V_{IH2}		12.5 V	Y1 Y2 Y3 Y4 Y5 Y6	11.25		11.25		11.25		
V_{OL1}	3007	39 40 41 42 43 44	5.5 V	I_{OL1}	V_{IL1}	I_{OL1}	V_{IL1}	I_{OL1}	V_{IL1}		V_{IL1}	I_{OL1}	V_{IL1}	I_{OL1}		V_{IL1}	I_{OL1}	5.5 V	Y1 Y2 Y3 Y4 Y5 Y6		0.5		0.5		0.5	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL2}	3007	45 46 47 48 49 50	5.0 V	I_{OL2}	V_{IL1}	I_{OL2}	V_{IL1}	I_{OL2}	V_{IL1}	GND	V_{IL1}	I_{OL2}	V_{IL1}	I_{OL2}		V_{IL1}	I_{OL2}	5.0 V	Y1 Y2 Y3 Y4 Y5 Y6		0.5		0.5		0.5	Vdc
V_{OL3}		51 52 53 54 55 56			V_{IL1}		V_{IL1}		V_{IL1}		V_{IL1}		V_{IL1}			V_{IL1}			Y1 Y2 Y3 Y4 Y5 Y6		0.05		0.05		0.05	
V_{OL4}		57 58 59 60 61 62	12.5 V		V_{IL2}		V_{IL2}		V_{IL2}		V_{IL2}		V_{IL2}			V_{IL2}		12.5 V	Y1 Y2 Y3 Y4 Y5 Y6		1.25		1.25		1.25	
I_{IH1} 6/	3010	63	15.0 V		15.0 V		15.0 V		15.0 V		15.0 V		15.0 V			15.0 V		15.0 V	All inputs together		6					nA
I_{IH2}		64 65 66 67 68 69			15.0 V GND		GND 15.0 V GND		GND GND 15.0 V GND		GND GND GND 15.0 V GND		GND 15.0 V GND			GND 15.0 V			A1 A2 A3 A4 A5 A6		1		45			
I_{IL1} 6/	3009	70														GND			All inputs together		-6					
I_{IL2}		71 72 73 74 75 76																	A1 A2 A3 A4 A5 A6		-1		-45			
																				Subgroup 4 T_C = 25°C						
																				Min	Max					
C_1	3012	77 78 79 80 81 82	GND		A 7/		A 7/		A 7/	GND	A 7/		A 7/			A 7/		GND	A1 A2 A3 A4 A5 A6		12					pF

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/ and 2/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 3/		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max8/	Min	Max8/	Min	Max8/	
t_{PHL}	3003 Fig. 5	83	5.0 V	OUT	IN	OUT	IN	OUT	IN	GND	IN	OUT	IN	OUT		IN	OUT	5.0 V	A1 to Y1	7/11	100/150	7/11	150/225	7/11	100/150	ns
		84																	A2 to Y2							
		85																	A3 to Y3							
		86																	A4 to Y4							
		87																	A5 to Y5							
		88																	A6 to Y6							
t_{PLH}		89		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		A1 to Y1	10/17	140/230	10/17	210/345	10/17	140/230	
		90																	A2 to Y2							
		91																	A3 to Y3							
		92																	A4 to Y4							
		93																	A5 to Y5							
		94																	A6 to Y6							
t_{THL}	3004 Fig. 5	95		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1	20/5	60/70	20/5	90/105	20/5	60/70	
		96																	Y2							
		97																	Y3							
		98																	Y4							
		99																	Y5							
		100																	Y6							
t_{TLH}		101		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1	20	270	20	405	20	270	
		102																	Y2							
		103																	Y3							
		104																	Y4							
		105																	Y5							
		106																	Y6							

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	For terminal conditions and limits see 1/ and 2/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4			1 mA			1 mA				1 mA			1 mA	GND	A1 A2 A3 A4		1.5					Vdc
$V_{IC(neg)}$		5 6 7 8			-1 mA			-1 mA	GND			-1 mA			-1 mA		A1 A2 A3 A4		-6					
I_{SS} I_{SS}	3005 See 5/	9 10			15.0 V GND			15.0 V GND				15.0 V GND			15.0 V GND	15.0 V 15.0 V	V_{SS} V_{SS}		-75 -75		-750 -750			nA nA
V_{OH1}	3006	11 12 13 14 15 16 17 18	I_{OH1}	I_{OH1}	V_{IH1} V_{IL1}	I_{OH1}	I_{OH1}	V_{IH1} V_{IL1}		I_{OH1}	I_{OH1}	V_{IH1} V_{IL1}	I_{OH1}	I_{OH1}	V_{IH1} V_{IL1}	4.5 V	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	2.5		2.5		2.5		Vdc
V_{OH2}		19 20 21 22 23 24 25 26	I_{OH2}	I_{OH2}	V_{IH1} V_{IL1}	I_{OH2}	I_{OH2}	V_{IH1} V_{IL1}		I_{OH2}	I_{OH2}	V_{IH1} V_{IL1}	I_{OH2}	I_{OH2}	V_{IH1} V_{IL1}	5.0 V	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	4.5		4.5		4.5		
V_{OH3}		27 28 29 30 31 32 33 34			V_{IH1} V_{IL1}			V_{IH1} V_{IL1}				V_{IH1} V_{IL1}			V_{IH1} V_{IL1}		Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	4.95		4.95		4.95		
V_{OH4}		35 36 37 38 39 40 41 42			V_{IH2} V_{IL2}			V_{IH2} V_{IL2}				V_{IH2} V_{IL2}			V_{IH2} V_{IL2}	12.5 V	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	11.25		11.25		11.25		

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	For terminal conditions and limits see 1/ and 2/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	43	I_{OL1}		V_{IL1}			V_{IL1}	GND							5.5 V	Y1		0.5		0.5		0.5	Vdc
		44				I_{OL1}				I_{OL1}		V_{IL1}			V_{IL1}		Y2							
		45											I_{OL1}				Y3							
		46															Y4							
		47		I_{OL1}	V_{IH1}												$\overline{Y1}$							
		48					I_{OL1}	V_{IH1}									$\overline{Y2}$							
		49									I_{OL1}	V_{IH1}					$\overline{Y3}$							
		50												I_{OL1}	V_{IH1}		$\overline{Y4}$							
V_{OL2}		51	I_{OL3}		V_{IL1}			V_{IL1}								5.0 V	Y1		0.5		0.5		0.5	
		52				I_{OL3}				I_{OL3}		V_{IL1}			V_{IL1}		Y2							
		53											I_{OL3}				Y3							
		54															Y4							
		55		I_{OL4}	V_{IH1}												$\overline{Y1}$							
		56					I_{OL4}	V_{IH1}									$\overline{Y2}$							
		57									I_{OL4}	V_{IH1}					$\overline{Y3}$							
		58												I_{OL4}	V_{IH1}		$\overline{Y4}$							
V_{OL3}		59			V_{IL1}			V_{IL1}									Y1		0.05		0.05		0.05	
		60										V_{IL1}			V_{IL1}		Y2							
		61															Y3							
		62															Y4							
		63			V_{IH1}												$\overline{Y1}$							
		64						V_{IH1}									$\overline{Y2}$							
		65										V_{IH1}					$\overline{Y3}$							
		66													V_{IH1}		$\overline{Y4}$							
V_{OL4}		67			V_{IL2}			V_{IL2}								12.5 V	Y1		1.25		1.25		1.25	
		68										V_{IL2}			V_{IL2}		Y2							
		69															Y3							
		70															Y4							
		71			V_{IH2}												$\overline{Y1}$							
		72						V_{IH2}									$\overline{Y2}$							
		73										V_{IH2}					$\overline{Y3}$							
		74													V_{IH2}		$\overline{Y4}$							
I_{IH1} 5/	3010	75			15.0 V			15.0 V				15.0 V			15.0 V	15.0 V	All inputs together		4					nA
I_{IH2}		76			15.0 V			GND				GND			GND		A1		1		45			
		77			GND			15.0 V				GND			GND		A2							
		78						GND				15.0 V			GND		A3							
		79						GND				GND			15.0 V		A4							
I_{IL1} 5/	3009	80						GND				GND			GND		All inputs together		-4					

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	For terminal conditions and limits see 1/ and 2/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_C = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	81 82 83 84			GND			GND	GND			GND			GND	15.0 V	A1 A2 A3 A4		-1		-45			nA
																		Subgroup 4 T_C = 25°C						
																		Min	Max					
C_I	3012	85 86 87 88			A 7/			A 7/	GND			A 7/			A 7/	GND	A1 A2 A3 A4		20					pF
																		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	89 90 91 92 93 94 95 96	OUT	OUT	IN IN	OUT	OUT	IN IN	GND	OUT	OUT	IN IN	OUT	OUT	IN IN	5.0 V	A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A1 to $\overline{Y1}$ A2 to $\overline{Y2}$ A3 to $\overline{Y3}$ A4 to $\overline{Y4}$	8	115	8	172	8	172	ns
t_{PLH}		97 98 99 100 101 102 103 104	OUT	OUT	IN IN	OUT	OUT	IN IN		OUT	OUT	IN IN	OUT	OUT	IN IN		A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A1 to $\overline{Y1}$ A2 to $\overline{Y2}$ A3 to $\overline{Y3}$ A4 to $\overline{Y4}$	9	125 110	9	188 165	9	188 165	
t_{THL}	3004 Fig. 5	105 106 107 108 109 110 111 112	OUT	OUT	IN IN	OUT	OUT	IN IN		OUT	OUT	IN IN	OUT OUT	IN	IN		A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A1 to $\overline{Y1}$ A2 to $\overline{Y2}$ A3 to $\overline{Y3}$ A4 to $\overline{Y4}$	8	50 110	8	75 165	8	75 165	
t_{TLH}		113 114 115 116 117 118 119 120	OUT	OUT	IN IN	OUT	OUT	IN IN		OUT	OUT	IN IN	OUT	OUT	IN IN		A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A1 to $\overline{Y1}$ A2 to $\overline{Y2}$ A3 to $\overline{Y3}$ A4 to $\overline{Y4}$	9	70 120	9	105 180	9	105 180	

See footnotes at end of device type 05.

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1/ Input pins not designated may be tied to V_{CC} or GND or may be left open provided they do not influence the outcome of the measurement. Output pins not designated may be tied to the loads or may be left open provided they do not influence the outcome of the measurement.

2/

Symbol	V_{IH1}		V_{IL1}		V_{IH2}		V_{IL2}		I_{OH1}	I_{OL1}	I_{OH2}	I_{OL2}	I_{OL3}	I_{OL4}
Device type	01 03 05	02 04	01 03 05	02 04	01 03 05	02 04	01 03 05	02 04	All	All	All	All	05	05
Temperature T_C = 25°C	3.95 V	3.8 V	0.9 V	1.1 V	10.25 V	9.5 V	2.15 V	2.8 V	-.1 mA	.23 mA	-.45 mA	3.0 mA	1.6 mA	0.8 mA
T_C = 125°C	3.85 V	3.6 V	0.65 V	0.85 V	10.0 V	9.25 V	1.95 V	2.55 V	-.1 mA	.23 mA	-.35 mA	2.1 mA	1.2 mA	0.55 mA
T_C = -55°C	4.05 V	3.95 V	0.95 V	1.35 V	10.5 V	9.75 V	2.24 V	3.05 V	-.1 mA	.23 mA	-.65 mA	3.7 mA	2.1 mA	1.0 mA

3/ Terminal 16 is not connected for device types 03 and 04.

4/ Test parameter $V_{IC(pos)}$ does not apply to device types 03 and 04.

5/ When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

6/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

7/ (A) Capacitance bridge between measured terminal and V_{SS}; frequency = 1 MHz.

8/ Test limits t_{PHL}, t_{PLH}, t_{THL}, and t_{TLH} for device types 01/03 and 02/04 consists of two sets of values and are expressed XXX/XXXX in the limits columns. The digits preceding the slash apply to the first device in a set.

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TABLE III. Group A inspection for device types 51 and 53.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$	3/	1 2 3 4 5 6	GND		1 mA		1 mA		1 mA		1 mA		1 mA			1 mA		GND	A1 A2 A3 A4 A5 A6		1.5					Vdc
$V_{IC(neg)}$		7 8 9 10 11 12			-1 mA		-1 mA		-1 mA	GND	-1 mA		-1 mA			-1 mA			A1 A2 A3 A4 A5 A6		-6					
I_{SS} I_{SS}	3005 4/ 3005 4/	13 14	18 V 18 V		18 V GND		18 V GND		18 V GND		18 V GND		18 V GND			18 V GND		18 V 18 V	V_{SS} V_{SS}		-75 -75		-750 -750			nAdc nAdc
V_{OH5}	3006	15 16 17 18 19 20	15 V		GND		GND		GND		GND		GND			GND		15 V	Y1 Y2 Y3 Y4 Y5 Y6	14.95		14.95		14.95		Vdc
V_{OL5}	3007	21 22 23 24 25 26	15 V		15 V		15 V		15 V		15 V		15 V			15 V			Y1 Y2 Y3 Y4 Y5 Y6		0.05		0.05		0.05	
V_{IH1}		27 28 29 30 31 32	5 V		4.0 V GND		GND 4.0 V GND		GND GND 4.0 V GND		GND GND GND 4.0 V GND		GND 4.0 V GND			GND 4.0 V		5 V	Y1 Y2 Y3 Y4 Y5 Y6		0.5		0.5		0.5	
V_{IH2}		33 34 35 36 37 38	10 V		8.0 V GND		8.0 V		8.0 V GND		8.0 V GND		8.0 V GND			GND 8.0 V		10 V	Y1 Y2 Y3 Y4 Y5 Y6		1.0		1.0		1.0	
V_{IH3}		39 40 41 42 43 44	15 V		12.5 V GND		GND 12.5 V GND		12.5 V GND GND GND		12.5 V GND GND		12.5 V GND			GND 12.5 V		15 V	Y1 Y2 Y3 Y4 Y5 Y6		1.5		1.5		1.5	

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TABLE III. Group A inspection for device types 51 and 53 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IL1}		45 46 47 48 49 50	5 V		1.0 V GND		GND 1.0 V GND		GND GND 1.0 V GND GND	GND	GND 1.0 V GND		GND 1.0 V GND			GND 1.0 V		5 V	Y1 Y2 Y3 Y4 Y5 Y6	4.5		4.5		4.5		Vdc
V_{IL2}		51 52 53 54 55 56	10 V		2.0 V GND		2.0 V GND		2.0 V GND		2.0 V GND		2.0 V GND			GND 2.0 V		10 V	Y1 Y2 Y3 Y4 Y5 Y6	9.0		9.0		9.0		
V_{IL3}		57 58 59 60 61 62	15 V		2.5 V GND		2.5 V GND		2.5 V GND		2.5 V GND		2.5 V GND			GND 2.5 V		15 V	Y1 Y2 Y3 Y4 Y5 Y6	13.5		13.5		13.5		
I_{OL1}		63 64 65 66 67 68	5 V	0.4 V	5 V GND	0.4 V	5 V GND	0.4 V	5 V GND		5 V GND	0.4 V	5 V GND	0.4 V		GND 5 V	0.4 V	5 V	Y1 Y2 Y3 Y4 Y5 Y6	3.2		2.2		4.0		mA
I_{OL2}		69 70 71 72 73 74	15 V	1.5 V	15 V GND	1.5 V	15 V GND	1.5 V	15 V GND		15 V GND	1.5 V	15 V GND	1.5 V		GND 15 V	1.5 V	15 V	Y1 Y2 Y3 Y4 Y5 Y6	24.0		17.0		30.0		
I_{OH1}		75 76 77 78 79 80	5 V	4.6 V		4.6 V		4.6 V				4.6 V		4.6 V		GND	4.6 V	5 V	Y1 Y2 Y3 Y4 Y5 Y6	3.2		2.2		4.0		
I_{OH2}		81 82 83 84 85 86	15 V	13.5 V		13.5 V		13.5 V				13.5 V		13.5 V			13.5 V	15 V	Y1 Y2 Y3 Y4 Y5 Y6	24.0		17.0		30.0		
I_{IH1} 5/	3010	87	18 V		18 V		18 V		18 V		18 V		18 V			18 V		18 V	All inputs together		6					nA

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TABLE III. Group A inspection for device types 51 and 53 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IH2}	3010	88	18 V		18 V		GND		GND	GND	GND		GND			GND		18 V	All		1		45			nA
		89			GND		18 V		GND		GND		GND			GND			A2							
		90					GND		18 V		18 V								A3							
		91							GND		GND								A4							
		92											18 V						A5							
		93											GND			18 V			A6							
I_{IL1} 5/	3009	94									GND		GND			GND			All inputs together		-6					
I_{IL2}		95			18 V		18 V		18 V		18 V		18 V			18 V			All		-1		-45			
		96					GND		18 V		18 V								A2							
		97					18 V		GND		18 V								A3							
		98							18 V		GND								A4							
		99									18 V		GND						A5							
		100									18 V		18 V			GND			A6							
																				Subgroup 4 T_C = 25°C						
																				Min	Max					
C_I	3012	101	GND		A 6/					GND								GND	All		20					pF
		102					A 6/												A2							
		103							A 6/										A3							
		104									A 6/								A4							
		105											A 6/						A5							
		106														A 6/			A6							

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TABLE III. Group A inspection for device types 51 and 53 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/	Measured terminal	Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max 7/	Min	Max 7/	Min	Max 7/	
t_{PHL}	3003 Fig. 5	107 108 109 110 111 112	5.0 V	OUT	IN	OUT	IN	OUT	IN	GND	IN	OUT	IN	OUT		IN	OUT	5.0 V	A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A5 to Y5 A6 to Y6	5/2	60/65	7/2	150/ 225	5/2	60/65	ns
t_{PLH}		113 114 115 116 117 118		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		A1 to Y1 A2 to Y2 A3 to Y3 A4 to Y4 A5 to Y5 A6 to Y6	7/8	140/ 120	10/8	210/ 345	7/8	140/ 120	
t_{THL}	3004 Fig. 5	119 120 121 122 123 124		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1 Y2 Y3 Y4 Y5 Y6	3/3	70/60	10/8	90/ 105	3/3	70/60	
t_{TLH}		125 126 127 128 129 130		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1 Y2 Y3 Y4 Y5 Y6	13	350/ 160	20	405	13	350/ 160	

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TABLE III. Group A inspection for device types 52 and 54.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$	3/	1 2 3 4 5 6	GND		1 mA		1 mA		1 mA		1 mA		1 mA			1 mA		GND	A1 A2 A3 A4 A5 A6		1.5					Vdc
$V_{IC(neg)}$		7 8 9 10 11 12			-1 mA		-1 mA		-1 mA	GND	-1 mA		-1 mA			-1 mA			A1 A2 A3 A4 A5 A6		-6					
I_{SS} I_{SS}	3005 4/ 3005 4/	13 14	18 V 18 V		18 V GND		18 V GND		18 V GND		18 V GND		18 V GND			18 V GND		18 V 18 V	V_{SS} V_{SS}		-75 -75		-750 -750			nAdc nAdc
V_{OH5}	3006	15 16 17 18 19 20	15 V		15 V		15 V		15 V		15 V		15 V			15 V		15 V	Y1 Y2 Y3 Y4 Y5 Y6	14.95		14.95		14.95		Vdc
V_{OL5}	3007	21 22 23 24 25 26	15 V		GND		GND		GND		GND		GND			GND			Y1 Y2 Y3 Y4 Y5 Y6		0.05		0.05		0.05	
V_{IH1}		27 28 29 30 31 32	5 V		3.5 V GND		GND 3.5 V GND		GND GND 3.5 V GND		GND GND GND 3.5 V GND		GND 3.5 V GND			GND 3.5 V		5 V	Y1 Y2 Y3 Y4 Y5 Y6	4.5		4.5		4.5		
V_{IH2}		33 34 35 36 37 38	10 V		7 V GND		7 V GND		7 V GND		7 V GND		7 V GND			GND 7 V		10 V	Y1 Y2 Y3 Y4 Y5 Y6	9.0		9.0		9.0		
V_{IH3}		39 40 41 42 43 44	15 V		11 V GND		GND 11 V GND		11 V GND GND GND		11 V GND GND		11 V GND			GND 11 V		15 V	Y1 Y2 Y3 Y4 Y5 Y6	13.5		13.5		13.5		

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TABLE III. Group A inspection for device types 52 and 54 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	Y_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IL1}		45 46 47 48 49 50	5 V		1.5 V GND		GND 1.5 V GND		GND GND 1.5 V GND	GND	GND GND GND 1.5 V GND		GND 1.5 V GND			GND 1.5 V		5 V	Y1 Y2 Y3 Y4 Y5 Y6		0.5		0.5		0.5	Vdc
V_{IL2}		51 52 53 54 55 56	10 V		3 V GND		3 V GND		3 V GND		3 V GND		3 V GND			GND 3 V		10 V	Y1 Y2 Y3 Y4 Y5 Y6		1.0		1.0		1.0	
V_{IL3}		57 58 59 60 61 62	15 V		4 V GND		4 V GND		4 V GND		4 V GND		4 V GND			GND 4 V		15 V	Y1 Y2 Y3 Y4 Y5 Y6		1.5		1.5		1.5	
I_{OL1}		63 64 65 66 67 68	5 V	0.4 V		0.4 V		0.4 V				0.4 V		0.4 V		GND	0.4 V	5 V	Y1 Y2 Y3 Y4 Y5 Y6	0.51		0.36		0.64		mA
I_{OL2}		69 70 71 72 73 74	15 V	1.5 V		1.5 V		1.5 V				1.5 V		1.5 V			1.5 V	15 V	Y1 Y2 Y3 Y4 Y5 Y6	3.4		2.4		4.2		
I_{OH1}		75 76 77 78 79 80	5 V	4.6 V	5 V GND	4.6 V	5 V GND	4.6 V	5 V GND		5 V GND	4.6 V	5 V GND	4.6 V		5 V	4.6 V	5 V	Y1 Y2 Y3 Y4 Y5 Y6	-0.51		-0.36		-0.64		
I_{OH2}		81 82 83 84 85 86	15 V	13.5 V	15 V GND	13.5 V	15 V GND	13.5 V	15 V GND GND GND		15 V GND GND	13.5 V	15 V GND	13.5 V		GND 15 V	13.5 V	15 V	Y1 Y2 Y3 Y4 Y5 Y6	-3.4		-2.4		-4.2		
I_{IH1} 5/	3010	87	18 V		18 V		18 V		18 V		18 V		18 V			18 V		18 V	All inputs together		6					nA

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TABLE III. Group A inspection for device types 52 and 54 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16 2/		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	V_{CC}	Y1	A1	Y2	A2	Y3	A3	V_{SS}	A4	Y4	A5	Y5	NC	A6	Y6	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IH2}	3010	88 89 90 91 92 93	18 V		18 V GND		GND 18 V GND		GND GND 18 Y GND	GND	GND GND GND 18 Y GND GND		GND 18 V GND			GND 18 V		18 V	A1 A2 A3 A4 A5 A6		1		45			nA
I_{IL1} 5/	3009	94									GND		GND			GND			All inputs together		-6					
I_{IL2}		95 96 97 98 99 100			18 V		18 V GND 18 V		18 V 18 V GND 18 Y		18 V 18 V 18 V GND 18 V 18 V		18 V GND 18 V			18 V GND			A1 A2 A3 A4 A5 A6		-1		-45			
																				Subgroup 4 T_C = 25°C						
																				Min	Max					
C_I	3012	101 102 103 104 105 106	GND		A 6/		A 6/		A 6/	GND	A 6/		A 6/			A 6/		GND	A1 A2 A3 A4 A5 A6		12					pF

See footnotes at end of device type 55.

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TABLE III. Group A inspection for device types 52 and 54 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z	For terminal conditions and limits see 1/																Measured terminal	Test limits: Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		Unit
		Test no.	1 V_{CC}	2 Y1	3 A1	4 Y2	5 A2	6 Y3	7 A3	8 V_{SS}	9 A4	10 Y4	11 A5	12 Y5	13 NC	14 A6	15 Y6	16 2/ V_{DD}		Min	Max 1/	Min	Max 1/	Min	Max 1/	
t_{PHL}	3003 Fig. 5	107	5.0 V	OUT	IN	OUT	IN	OUT	IN	GND	IN	OUT	IN	OUT		IN	OUT	5.0 V	A1 to Y1	5/7	60/65	7/11	150/225	5/7	60/65	ns
		108																	A2 to Y2							
		109																	A3 to Y3							
		110																	A4 to Y4							
		111																	A5 to Y5							
		112																	A6 to Y6							
t_{PLH}		113		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		A1 to Y1	7/11	140/120	10/17	210/345	7/11	140/120	
		114																	A2 to Y2							
		115																	A3 to Y3							
		116																	A4 to Y4							
		117																	A5 to Y5							
		118																	A6 to Y6							
t_{THL}	3004 Fig. 5	119		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1	3	70/60	4/5	90/105	3	70/60	
		120																	Y2							
		121																	Y3							
		122																	Y4							
		123																	Y5							
		124																	Y6							
t_{TLH}		125		OUT	IN	OUT	IN	OUT	IN		IN	OUT	IN	OUT		IN	OUT		Y1	13	350/160	20	405	13	350/160	
		126																	Y2							
		127																	Y3							
		128																	Y4							
		129																	Y5							
		130																	Y6							

See footnotes at end of device type 55.

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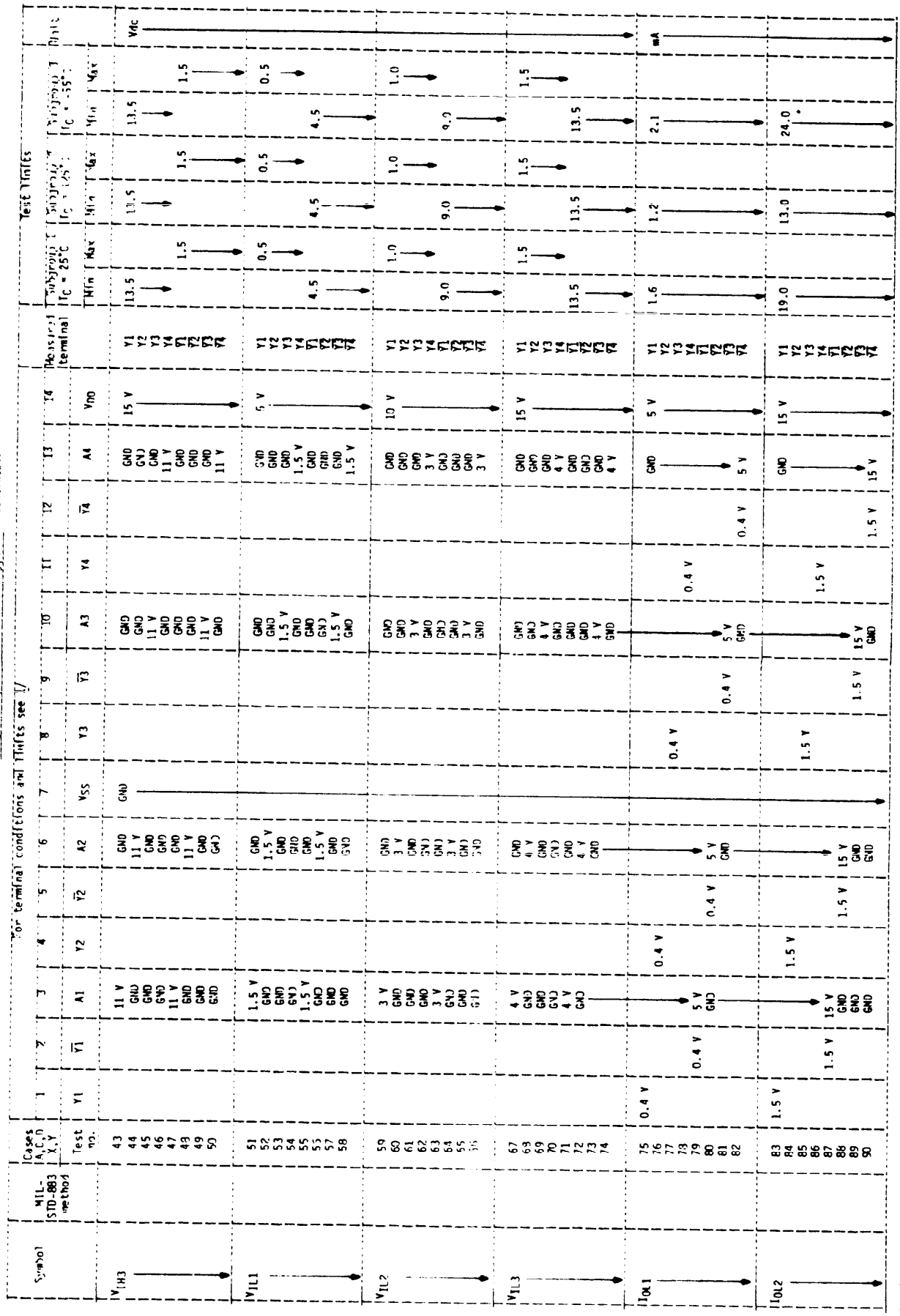
TABLE III. Group A inspection for device type 55.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	For terminal conditions and limits see 1/															Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14			Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}			Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1			1 mA											GND		A1		1.5					Vdc
		2						1 mA										A2							
		3										1 mA						A3							
		4													1 mA			A4							
$V_{IC(neg)}$		5			-1 mA				GND									A1		-6					
		6						-1 mA										A2							
		7										-1 mA						A3							
		8													-1 mA			A4							
I_{SS}	3005 4/	9			15 V			15 V				15 V			15 V	15 V		V_{SS}		-75		-750			nA
I_{SS}	3005 $\overline{4}$/	10			GND			GND				GND			GND			V_{SS}		-75		-750			nA
V_{OH5}	3006	11			15 V													Y1	14.95		14.95		14.95		Vdc
		12						15 V										Y2							
		13										15 V						Y3							
		14													15 V			Y4							
		15			GND													$\overline{Y1}$							
		16						GND										$\overline{Y2}$							
		17										GND						$\overline{Y3}$							
		18													GND			$\overline{Y4}$							
V_{OL5}	3007	19			GND													Y1		0.05		0.05		0.05	
		20						GND										Y2							
		21										GND						Y3							
		22													GND			Y4							
		23			15 V													$\overline{Y1}$							
		24						15 V										$\overline{Y2}$							
		25										15 V						$\overline{Y3}$							
		26													15 V			$\overline{Y4}$							
V_{IH1}		27			3.5 V			GND				GND			GND	5 V		Y1	4.5		4.5		4.5		
		28			GND			3.5 V				GND			GND			Y2							
		29			GND			GND				3.5 V			GND			Y3							
		30			GND			GND				GND			3.5 V			Y4							
		31			3.5 V			GND				GND			GND			$\overline{Y1}$		0.5		0.5		0.5	
		32			GND			3.5 V				GND			GND			$\overline{Y2}$							
		33			GND			GND				3.5 V			GND			$\overline{Y3}$							
		34			GND			GND				GND			3.5 V			$\overline{Y4}$							
V_{IH2}		35			7 V			GND				GND			GND	10 V		Y1	9.0		9.0		9.0		
		36			GND			7 V				GND			GND			Y2							
		37			GND			GND				7 V			GND			Y3							
		38			GND			GND				GND			7 V			Y4							
		39			7 V			GND				GND			GND			$\overline{Y1}$		1.0		1.0		1.0	
		40			GND			7 V				GND			GND			$\overline{Y2}$							
		41			GND			GND				GND			7 V			$\overline{Y3}$							
		42			GND			GND				GND			7 V			$\overline{Y4}$							

See footnotes at end of device type 55.

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TABLE III. Group A inspection for device type 55 - Continued.



Symbol	MIL-STD-883 method	Cases A, C, D X, Y	Test no.	For terminal conditions and limits see 1/														Test limits				Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Subgroups 1, 2, 3 T_C = 25°C		Subgroups 1, 2, 3 T_C = -55°C, +125°C	
																			Min	Max	Min	Max	
V_{IH3}			43	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	13.5		13.5		Vdc
			44			11 V			11 V	GND			GND			GND	15 V					
			45			GND			GND				GND			GND							
			46			GND			GND				11 V			GND							
			47			11 V			GND				GND			GND							
			48			GND			GND				GND			GND							
			49			GND			GND				GND			GND							
			50			GND			GND				11 V			11 V							
V_{IL1}			51			1.5 V			GND				GND			GND							
			52			GND			1.5 V				GND			GND							
			53			GND			GND				1.5 V			GND							
			54			GND			GND				GND			GND							
			55			1.5 V			GND				GND			GND							
			56			GND			1.5 V				GND			GND							
			57			GND			GND				GND			GND							
			58			GND			GND				1.5 V			1.5 V							
V_{IL2}			59			3 V			GND				GND			GND							
			60			GND			3 V				GND			GND							
			61			GND			GND				3 V			GND							
			62			GND			GND				GND			GND							
			63			3 V			GND				GND			GND							
			64			GND			3 V				GND			GND							
			65			GND			GND				GND			GND							
			66			GND			GND				3 V			3 V							
V_{IL3}			67			4 V			GND				GND			GND							
			68			GND			4 V				GND			GND							
			69			GND			GND				4 V			GND							
			70			GND			GND				GND			GND							
			71			4 V			GND				GND			GND							
			72			GND			4 V				GND			GND							
			73			GND			GND				4 V			GND							
			74			GND			GND				GND			GND							
I_{OL1}			75	0.4 V			0.4 V				0.4 V			0.4 V			5 V		1.6		1.2		mA
			76			5 V		0.4 V	5 V			0.4 V	5 V		0.4 V	GND							
			77		0.4 V	GND			GND				GND			5 V							
			78																			
			79																			
			80																			
			81																			
			82																			
I_{OL2}			83	1.5 V			1.5 V				1.5 V			1.5 V			15 V		19.0		13.0		mA
			84			15 V		1.5 V	15 V			1.5 V	15 V		1.5 V	GND							
			85		1.5 V	GND			GND				GND			15 V							
			86																			
			87																			
			88																			
			89																			
			90																			

See footnotes at end of device type 55.

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TABLE III. Group A inspection for device type 55 - Continued.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	Test no.	For terminal conditions and limits see 1/														Test limits						Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C	
																			Min	Max	Min	Max	Min	Max	
I_{OH1}			91	Y1 4.6 V	$\overline{Y1}$	A1 5 V GND	Y2 4.5 V	$\overline{Y2}$	A2 GND 5 V	Y_{SS} GND	Y3 4.6 V	$\overline{Y3}$ 4.6 V	A3 GND 5 V GND	r4 4.6 V	$\overline{Y4}$	A4 GND GND GND 5 V GND	V_{DD} 5 V	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	-1.6		-1.2		-2.1		nA
			92																					
			93		4.5 V																			
			94																					
			95					4.6 V	GND															
			96																					
			97												4.6 V									
			98																					
I_{OH2}			99	13.5 V		15 V GND	13.5 V		15 V GND		13.5 V		15 V GND	13.5 V		15 V GND	15 V	Y1 Y2 Y3 Y4 $\overline{Y1}$ $\overline{Y2}$ $\overline{Y3}$ $\overline{Y4}$	-19.0		-13.0		-24.0		
			100																					
			101																					
			102																					
			103		13.5 V																			
			104																					
			105					13.5 V				13.5 V												
			106																					
I_{IH1} 5/	3010		107			18 V			18 V				18 V			18 V	18 V	All inputs together	4					mA
I_{IH2}			108			18 V GND			GND 18 V GND GND				GND GND 18 V GND			GND GND GND 18 V		A1 A2 A3 A4	1	45				
			109																					
			110																					
			111																					
I_{IL1} 5/	3009		112						GND				GND			GND		All inputs together	-4					
I_{IL2}			113			GND			18 V GND				18 V GND			18 V 18 V GND		A1 A2 A3 A4	-1	-45				
			114			18 V																		
			115			18 V			18 V				GND 18 V			18 V								
			116			18 V																		
																			Subgroups 4 T_C = 25°C					
																			Min	Max				
C_I	3012		117			A 6/			A 6/	GND			A 6/			A 6/	GND	A1 A2 A3 A4		12				pF
			118																					
			119																					
			120																					

See footnotes at end of device type 55.

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TABLE III. Group A inspection for device type 55 - Continued.

Symbol	MIL-STD-883 method	Cases A,C,D X,Y	For terminal conditions and limits see 1/														Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	Y1	$\overline{Y1}$	A1	Y2	$\overline{Y2}$	A2	V_{SS}	Y3	$\overline{Y3}$	A3	Y4	$\overline{Y4}$	A4	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	121	OUT		IN				GND							5.0 V	A1 to Y1	5	115	8	172	5	115	ns
		122				OUT		IN									A2 to Y2							
		123								OUT		IN					A3 to Y3							
		124											OUT		IN		A4 to Y4							
		125		OUT	IN												A1 to $\overline{Y1}$							
		126					OUT	IN									A2 to $\overline{Y2}$							
		127									OUT	IN					A3 to $\overline{Y3}$							
		128												OUT	IN		A4 to $\overline{Y4}$							
t_{PLH}		129	OUT		IN												A1 to Y1	6	125	9	188	6	125	
		130				OUT		IN									A2 to Y2							
		131								OUT		IN					A3 to Y3							
		132											OUT		IN		A4 to Y4							
		133		OUT	IN												A1 to $\overline{Y1}$	5	110	8	165	5	110	
		134					OUT	IN									A2 to $\overline{Y2}$							
		135									OUT	IN					A3 to $\overline{Y3}$							
		136												OUT	IN		A4 to $\overline{Y4}$							
t_{THL}	3004 Fig. 5	137	OUT		IN												A1 to Y1	2	50	3	75	2	50	
		138				OUT		IN									A2 to Y2							
		139								OUT		IN					A3 to Y3							
		140											OUT		IN		A4 to Y4							
		141		OUT	IN												A1 to $\overline{Y1}$	5	110	8	165	5	110	
		142					OUT	IN									A2 to $\overline{Y2}$							
		143									OUT	IN					A3 to $\overline{Y3}$							
		144												OUT	IN		A4 to $\overline{Y4}$							
t_{TLH}		145	OUT		IN												A1 to Y1	3	70	5	105	3	70	
		146				OUT		IN									A2 to Y2							
		147								OUT		IN					A3 to Y3							
		148											OUT		IN		A4 to Y4							
		149		OUT	IN												A1 to $\overline{Y1}$	6	120	9	180	6	120	
		150					OUT	IN									A2 to $\overline{Y2}$							
		151									OUT	IN					A3 to $\overline{Y3}$							
		152												OUT	IN		A4 to $\overline{Y4}$							

1/ Input pins not designated may be tied to V_{CC} or GND or may be left open provided they do not influence the outcome of the measurement. Output pins not designated may be tied to the loads or may be left open provided they do not influence the outcome of the measurement.

2/ Terminal 16 is not connected for device types 53 and 54.

3/ Test parameter $V_{IC(pos)}$ does not apply to device types 53 and 54.

4/ When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

5/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

6/ (A) Capacitance bridge between measured terminal and V_{SS}; frequency = 1 MHz.

7/ Test limits t_{PHL}, t_{PLH}, t_{THL}, and t_{TLH} for device types 51/53 and 52/54 consists of two sets of values and are expressed XXX/XXXX in the limits columns. The digits preceding the slash apply to the first device in a set.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection, and shall consist of tests specified in table IV herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

1. Test condition D and as specified in 4.5.2 and as shown on figure 4 or equivalent.

2. T_A = 125°C, minimum.

3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device type	
	01-05	51-55
I_{SS}	±20 nA	±20 nA
V_{OL1}	±.04 V	---
V_{OH1}	±.08 V	---
I_{OL1}	---	±15%
I_{OH1}	---	±15%

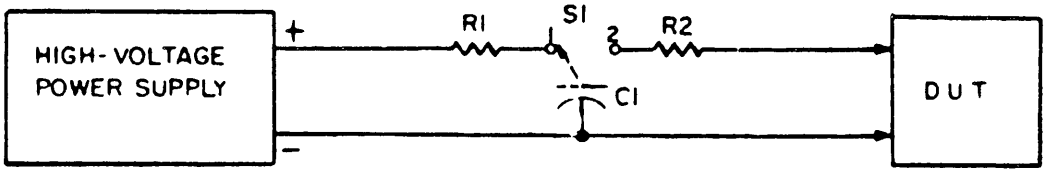
1/ Each parameter shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

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4.5.2 Burn-in and life test cool down procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to an ambient temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 6:

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



V_{ZAP} = 400 V charge on C1.

1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 6. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IH}, and I_{IL} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

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4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 7. In situ and remote testing, the test shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	01-05	51-55
V_{TN}	0.3 V min	V_{DD} = 10 V	V_{DD} = 10 V
V_{TP}	2.8 V max	V_{DD} = 10 V	V_{DD} = 10 V
ΔV_T	1.4 V max	V_{DD} = 10 V	V_{DD} = 10 V
I_{SS}	100 x max limit	V_{DD} = 15 V	V_{DD} = 18 V
t_{PLH}	1.35 x max limit	V_{DD} = 5 V	V_{DD} = 5 V
t_{PHL}	1.35 x max limit	V_{DD} = 5 V	V_{DD} = 5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	10 V dc through a 30 kΩ to 60 kΩ resistor	V_{SS} = GND	V_{CC} and V_{DD} = 10 V dc
01, 51	3, 5, 7, 9, 11, 14	8	1, 16
02, 52	3, 5, 7, 9, 11, 14	8	1, 16
03, 53	3, 5, 7, 9, 11, 14	8	1
04, 54	3, 5, 7, 9, 11, 14	8	1
05, 55	3, 6, 10, 13	7	14

Pins not designated are open or connected to 10 V dc through a 30 kΩ to 60 kΩ resistor.

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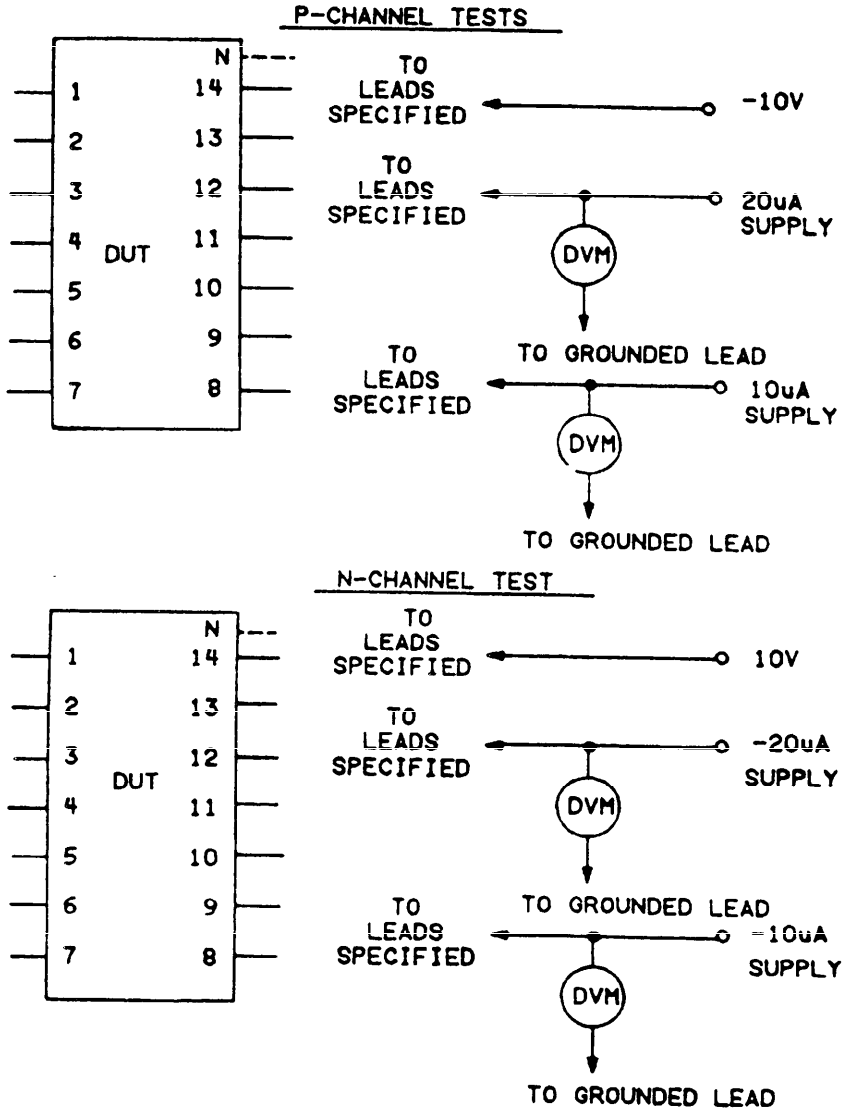


FIGURE 7. Threshold-voltage test circuit.

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TABLE VIII. Threshold voltage test circuit conditions.

Device type	GND	10 V	V_{TN} measured at		GND	-10 V	V_{TP} measured at	
			-20 µA supply	-10 µA supply			20 µA supply	10 µA supply
01, 51	3	1, 16		5, 7, 8, 9, 11, 14	3	5, 7, 8, 9, 11, 14		1, 16
02, 52	3	1, 16		5, 7, 8, 9, 11, 14	3	5, 7, 8, 9, 11, 14		1, 16
03, 53	3	1		5, 7, 8, 9, 11, 14	3	5, 7, 8, 9, 11, 14		1
04, 54	3	1		5, 7, 8, 9, 11, 14	3	5, 7, 8, 9, 11, 14		1
05, 55	3	14		6, 7, 10, 13	3	6, 7, 10, 13		14

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

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f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirement for "JAN" marking.

i. Requirement for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{ZAP}	Input test voltage.
C_i	Input terminal-to-V_{SS} capacitance.
GND	Ground. Zero voltage potential.
T_A	Free air temperature.
t_{THL}	Fall time. Time duration during which the amplitude of the trailing edge of the input forcing condition or waveform is decreasing from 90 to 10 percent of the maximum amplitude.
t_{TLH}	Rise time. Time duration during which the amplitude of the leading edge of the input forcing condition or waveform is increasing from 10 to 90 percent of the maximum amplitude.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
I_{SS}	Quiescent supply current.
$V_{IC(pos)}$	Positive clamping input to V_{DD}.
$V_{IC(neg)}$	Negative clamping input to V_{SS}.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support shall be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

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In all applications except multiplexers, four of the military device types are preferred for new designs and substitutable as follows. Device types 03 and 04 are replacements for device types 01 and 02 respectively. Device types 53 and 54 are replacements for device types 51 and 52 respectively.

Military device type	Generic-industry type
01	4009A
02	4010A
03	4049A
04	4050A
05	4041A
51	4009UB
52	4010B
53	4049UB
54	4050B
55	4041UB

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, OS, CG, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/55F	Microcircuits, Digital, CMOS, Buffer/Converter, True/Complement Buffer, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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